# Q2A | Summarize the Data

**Class:** CS P556 - Applied Machine Learning
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**Data:** the daily file of the Bike Sharing Dataset from the UCI Machine Learning Repository.

**Disclosure:** This notebook was completed with the assistance of Claude Code (Opus 5.5) and the book *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd edition) by Aurélien Géron.

> A. Summarize the data. How much data is present? What attributes/features are continuous valued? Which attributes are categorical?

This notebook answers the question in three parts:

1. **How much data is present?** Rows, columns, missing values, and duplicates.
2. **Which attributes are continuous?** Data types, distinct values, value ranges, and the date column.
3. **Which attributes are categorical?** The values each categorical attribute takes, and the one column that is neither.

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**IMPORTANT**

The dataset is a single file with no `train` / `test` split. The split is made in a later notebook, so this summary describes the whole dataset.

# Answer Summary

- **How much data is present?** 731 rows and 16 columns (the target `Rentals` and 15 others), with no missing values and no duplicate rows. Each row is one day, from 2011-01-01 to 2012-12-31.
- **Which attributes are continuous?** 8: `Temp`, `FeelsLike`, `Humidity`, and `Wind` (decimals between 0 and 1), `Casual`, `Registered`, and `Rentals` (whole-number counts with 606 to 696 distinct values), and `Date` (it measures time, so the gap between two dates is an amount of days).
- **Which attributes are categorical?** 7: `Season`, `Year`, `Month`, `IsHoliday`, `Weekday`, `IsWorkday`, and `Weather`, each with 2 to 12 distinct values. `ID` is neither: it is a label for the row.

**Flags:**

- The rows are in date order, one per day with no gaps, which matters for how the data is split.
- `Date` is not stored as a number, and `ID` carries the same information as `Date`.
- `Casual` and `Registered` add up to `Rentals`, which matters for whether they can be used as attributes.
- The categorical attributes are stored as integer codes, and `IsHoliday` and `Weather` each have a value with only 21 rows.
- The scales differ: 0 to 1 for the measurements, up to 12 for the codes, thousands for the counts.

- `Humidity` is the only decimal column with a minimum of 0, a value to check as a possible error.

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# Imports

In [1]:
```
import pandas as pd
from IPython.display import display

import Q2.consts as Q2_CONSTS

from utils import read_csv_to_df
```

# Fetch Data

In [2]:
```
df = read_csv_to_df(path=Q2_CONSTS.DATA_PATH, parse_dates=["dteday"])
df = df.rename(columns=Q2_CONSTS.SHORT_NAMES)
```

# 1. How Much Data is Present?

In [3]:
```
summary = pd.DataFrame(
        {
                "row count": [ df.shape[0] ],
                "columns count": [ df.shape[1] ],
                "missing values": [ df.isnull().sum().sum() ],
                "rows w/ missing values": [ df.isnull().any(axis=1).sum() ],
                "% rows w/ missing values": [
                        round(df.isnull().any(axis=1).sum() / df.shape[0] * 100, 2),
                ]
        },
        index=["Full Dataset"],
        dtype=object,  # keeps the counts as whole numbers after the transpose
)

summary.T
```

Out[3]:

| | Full Dataset |
|---|---|
| **row count** | 731 |
| **columns count** | 16 |
| **missing values** | 0 |
| **rows w/ missing values** | 0 |
| **% rows w/ missing values** | 0.0 |

- 16 columns: the target `Rentals` and 15 others. No value is missing, so there is nothing to impute.

## Duplicate Rows

In [4]:
```
df.duplicated().sum()
```

Out[4]:
```
0
```

- No duplicate rows, so there is nothing to remove.

# 2. Which Attributes are Continuous?

An attribute is continuous if it measures an amount that can take many values within a range. The stored data type does not settle this, so the evidence is the number of distinct values and the range of each column.

In [5]:
```
info_df = pd.DataFrame({
        "dtype": df.dtypes,
        "unique values": df.nunique(),
        "most frequent value count": df.apply(lambda col: col.value_counts().iloc[0]),
}).sort_values(by=["unique values", "most frequent value count"], ascending=False)

info_df
```

Out[5]:

| | dtype | unique values | most frequent value count |
|---|---|---|---|
| **ID** | int64 | 731 | 1 |
| **Date** | datetime64[ns] | 731 | 1 |
| **Rentals** | int64 | 696 | 2 |
| **FeelsLike** | float64 | 690 | 4 |
| **Registered** | int64 | 679 | 3 |
| **Wind** | float64 | 650 | 3 |
| **Casual** | int64 | 606 | 4 |
| **Humidity** | float64 | 595 | 4 |
| **Temp** | float64 | 499 | 5 |
| **Month** | int64 | 12 | 62 |
| **Weekday** | int64 | 7 | 105 |
| **Season** | int64 | 4 | 188 |
| **Weather** | int64 | 3 | 463 |
| **IsHoliday** | int64 | 2 | 710 |
| **IsWorkday** | int64 | 2 | 500 |
| **Year** | int64 | 2 | 366 |

- The distinct value counts split the 16 columns into two groups with nothing in between: 7 columns with 2 to 12 distinct values, and 9 columns with 499 to 731 where no value repeats more than 5 times.
- All 7 columns in the first group are `int64`, like the counts `Casual`, `Registered`, and `Rentals`, so the data type does not tell the two groups apart.
- `Date` is the only non-numeric column (`datetime64`).

## Value Ranges

In [6]:
```
numeric_df = df.select_dtypes(include="number")

value_ranges = pd.DataFrame({
        # object dtype keeps whole-number columns without decimals
```

```
        "min": pd.Series({col: round(numeric_df[col].min(), 3) for col in numeric_df},
dtype=object),
        "max": pd.Series({col: round(numeric_df[col].max(), 3) for col in numeric_df},
dtype=object),
        "mean": numeric_df.mean().round(3),
        "std": numeric_df.std().round(3),
        "whole numbers only": numeric_df.apply(lambda col: (col.dropna() % 1 ==
0).all()),
}, index=numeric_df.columns)
display(value_ranges)

pd.DataFrame(
        {
                "rows where Casual + Registered = Rentals": [ (df["Casual"] +
df["Registered"] == df["Rentals"]).sum() ],
        },
        index=["Full Dataset"],
).T
```

| | min | max | mean | std | whole numbers only |
|---|---|---|---|---|---|
| **ID** | 1 | 731 | 366.000 | 211.166 | True |
| **Season** | 1 | 4 | 2.497 | 1.111 | True |
| **Year** | 0 | 1 | 0.501 | 0.500 | True |
| **Month** | 1 | 12 | 6.520 | 3.452 | True |
| **IsHoliday** | 0 | 1 | 0.029 | 0.167 | True |
| **Weekday** | 0 | 6 | 2.997 | 2.005 | True |
| **IsWorkday** | 0 | 1 | 0.684 | 0.465 | True |
| **Weather** | 1 | 3 | 1.395 | 0.545 | True |
| **Temp** | 0.059 | 0.862 | 0.495 | 0.183 | False |
| **FeelsLike** | 0.079 | 0.841 | 0.474 | 0.163 | False |
| **Humidity** | 0.0 | 0.972 | 0.628 | 0.142 | False |
| **Wind** | 0.022 | 0.507 | 0.190 | 0.077 | False |
| **Casual** | 2 | 3410 | 848.176 | 686.622 | True |
| **Registered** | 20 | 6946 | 3656.172 | 1560.256 | True |
| **Rentals** | 22 | 8714 | 4504.349 | 1937.211 | True |

Out[6]:

| | Full Dataset |
|---|---|
| **rows where Casual + Registered = Rentals** | 731 |

- The scales differ, which matters for SGD and regularization: the 4 decimal columns lie between 0 and 1, while the counts reach the thousands.
- `Casual` and `Registered` add up to `Rentals` in all 731 rows (the check table right above), as two parts of the target would. This matters for whether they can be used as attributes.
- `Humidity` is the only decimal column with a minimum of 0, a value to check as a possible error.

7 of the 15 numeric columns are continuous: the 4 decimal columns (measurements) and the 3 rental counts, which are whole numbers but are amounts with 606 to 696 distinct values.

## Date

In [7]:
```
days_since_first = (df["Date"] - df["Date"].min()).dt.days

date_summary = pd.DataFrame(
        {
                "first date": [ df["Date"].min().date() ],
                "last date": [ df["Date"].max().date() ],
                "distinct dates": [ df["Date"].nunique() ],
                "days from first to last date": [ days_since_first.max() ],
                "rows 1 day after the previous row": [ (df["Date"].diff().dt.days ==
1).sum() ],
                "rows where ID = days since first date + 1": [ (df["ID"] ==
days_since_first + 1).sum() ],
        },
        index=["Date"],
)

date_summary.T
```

Out[7]:

| | Date |
|---|---|
| **first date** | 2011-01-01 |
| **last date** | 2012-12-31 |
| **distinct dates** | 731 |
| **days from first to last date** | 730 |
| **rows 1 day after the previous row** | 730 |
| **rows where ID = days since first date + 1** | 731 |

- The rows are in date order, one per day with no gaps (all 730 rows after the first are 1 day after the row before), which matters for how the data is split.
- `Date` measures time, so it is continuous: counted as days since the first date it is an amount from 0 to 730, and the gap between two dates means something.
- `ID` is a label for the row, so it is neither continuous nor categorical. It equals that day count plus 1 in all 731 rows only because the file is in date order with no gaps, so it carries the same information as `Date`.

**8 attributes are continuous: the 4 measurements (`Temp`, `FeelsLike`, `Humidity`, and `Wind`), the 3 rental counts (`Casual`, `Registered`, and the target `Rentals`), and `Date`.**

# 3. Which Attributes are Categorical?

An attribute is categorical if its values name a group that a row belongs to: the value is a label, not an amount. The candidates are the 7 columns with 12 or fewer distinct values from section 2.

In [8]:
```
categorical_cols = ["Season", "Year", "Month", "IsHoliday", "Weekday", "IsWorkday",
"Weather"]

category_levels = pd.DataFrame({
        "distinct values": df[categorical_cols].nunique(),
        "values": [", ".join(map(str, sorted(df[col].unique()))) for col in
categorical_cols],
        "fewest rows per value": [df[col].value_counts().min() for col in
categorical_cols],
        "most rows per value": [df[col].value_counts().max() for col in
categorical_cols],
})
```

```
category_levels
```

Out[8]:

| | distinct values | values | fewest rows per value | most rows per value |
|---|---|---|---|---|
| **Season** | 4 | 1, 2, 3, 4 | 178 | 188 |
| **Year** | 2 | 0, 1 | 365 | 366 |
| **Month** | 12 | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 | 57 | 62 |
| **IsHoliday** | 2 | 0, 1 | 21 | 710 |
| **Weekday** | 7 | 0, 1, 2, 3, 4, 5, 6 | 104 | 105 |
| **IsWorkday** | 2 | 0, 1 | 231 | 500 |
| **Weather** | 3 | 1, 2, 3 | 21 | 463 |

- All 7 take a few whole-number values, and their names describe groups of days, so the numbers are labels, not amounts. They are stored as integer codes, which matters for how they are encoded.
- `IsHoliday` and `Weather` each have a value with only 21 of the 731 rows, which matters for any split of the rows.

**7 attributes are categorical: `Season`, `Year`, `Month`, `IsHoliday`, `Weekday`, `IsWorkday`, and `Weather`.**

**1 column is neither continuous nor categorical: `ID`, the row number (see the Date section).**